3D Immersed-Boundary Cylinder: Preliminary Findings

3dRotatingCylinder Project

October 29, 2026

1 Overview

These simulations model a rigid circular cylinder, free to translate in a viscous fluid under gravity, while being driven by a prescribed oscillatory cross-flow velocity. The cylinder is extruded along z with periodic boundary conditions in that direction, so the flow is effectively three-dimensional only through whatever spanwise structure the body introduces. The question this preliminary set of runs addresses is simple: *does adding radial fins along the cylinder's span change how it glides through the fluid, relative to a bare cylinder?*

Four cases form a matched comparison, differing only in the number of fins (0, 1, 2, or 3) attached at evenly spaced points along the span, with identical fluid properties, domain, grid, and forcing: `faster_nofin_full_span`, `faster_midfin`, `faster_2fin`, and `faster_3fin`. Two earlier cases, `faster_nofin` and `faster_nofin_large`, established the baseline bare-cylinder behavior and motivated the larger domain used by the fin series. A separate case, `endcaps`, tests end discs rather than fins and uses different flow parameters, so it is not part of the controlled comparison.

All reported trajectories, glide angles, and status data are transferred from the hand-written `README.md` in each case's directory and from the project's case index; see the project repository for the full narrative and regeneration instructions.

2 Case setups

2.1 Common parameters (fin series)

`faster_nofin_full_span`, `faster_midfin`, `faster_2fin`, and `faster_3fin` share the following setup:

- **Body:** solid circular cylinder, radius $R = 3.17$ (diameter $D = 6.34$), axis along z, length $L = 35.0$ (spans the full periodic z-domain).
- **Fluid:** $\rho = 1.0$, $\mu = 0.01$, giving $\mathrm{Re} = \rho U D/\mu \approx 634$ based on diameter.
- **Domain:** $x \in [-30, 120]$, $y \in [-120, 30]$, $z \in [-17.5, 17.5]$; periodic in z.
- **Grid:** $150 \times 150 \times 35$ base grid, 4 AMR levels (refinement $4 \cdot 4 \cdot 2$), finest spacing $\Delta x = 0.0625$.
- **Forcing:** prescribed cross-flow velocity $U_\infty \cos(2\pi f t)$, $U = 1.0$, $f = 0.6$ (Keulegan–Carpenter number $U/(fD) \approx 0.26$).
- **Buoyancy:** neutrally buoyant body ($\rho_{\text{solid}} = \rho_{\text{fluid}}$), net body force from a reduced gravity term ($\delta = 0.011$) active from $t = 0$, giving the cylinder a downward glide as it oscillates.

- **Run length:** target END_TIME = 30 (some cases were restarted past this; see Section 3).

The immersed-boundary point cloud and the Eulerian grid are tied together: the point spacing implied by each case's body parameters must equal the finest Eulerian grid spacing above ($\Delta x = 0.0625$), or the one-point- per-finest-cell assumption the IB method relies on is violated. This is a modeling invariant across every case in this report, not something tuned per case.

The cases differ only in how many fins are attached and where:

Case	Fins	z-positions (fraction of span)	IB points
`faster_nofin_full_span`	0	—	4,524,800
`faster_midfin`	1	1/2	4,549,032
`faster_2fin`	2	1/4, 3/4	4,573,264
`faster_3fin`	3	1/6, 1/2, 5/6	4,597,496

Table 1: The fin series. Each fin has radius $2R = 6.34$, i.e. twice the cylinder's own radius.

`faster_midfin` is the single-fin case: one fin at the axial midpoint. `faster_2fin` and `faster_3fin` add fins symmetrically outward from there.

2.2 Baseline series

`faster_nofin` is the original bare-cylinder baseline: the same cylinder (but shorter, $L = 25.5$, non-periodic z) in a smaller domain ($x \in [-20, 80]$, $y \in [-72, 18]$). Its first run exited the domain at $t = 21.55$ when the body's center of mass approached the lower y boundary. `faster_nofin_large` repeats it with the domain enlarged to $x \in [-30, 120]$, $y \in [-120, 30]$ (the same size later reused by the fin series) and completed the full run. The fin series uses this larger domain, plus a longer, periodic-z body so the cylinder is uniform across its whole span.

2.3 `endcaps` (not part of the controlled comparison)

`endcaps` adds two discs (radius 7.0) capping the ends of a `faster_nofin`-sized cylinder, with no fins along the span. It also changes the oscillation frequency ($f = 0.333$ vs. 0.6), gravity onset ($t = 3.0$ vs. $t = 0$), and the upper-y boundary condition (closed vs. open) relative to `faster_nofin`, so it cannot be read as an isolated effect of end geometry.

3 Findings

3.1 Run status

All three finned cases stall in the same place: every one of `faster_midfin`, `faster_2fin`, and `faster_3fin` reaches $x \approx 110$–113, close to the domain's $x_{\max} = 120$, before crashing or timing out. The two crashes are the identical failure mode (an IB spreading operator running out of ghost cells once the body gets close enough to the physical boundary); the timeout case is still drifting toward the same region when its wall-clock budget runs out. This is consistent across all three finned cases and looks like a boundary proximity problem rather than anything specific to a given fin count — the bare-cylinder case in the same domain, `faster_nofin_full_span`, reaches almost the same x (112.5) but completes cleanly, which narrows the likely cause to the fins' wider IB stencil needing more ghost-cell width near the wall, not the boundary width itself.

Case	Status	t_{end}	Final COM (x, y)
`faster_nofin`	domain exit	21.5	$(63.9, -67.5)$
`faster_nofin_large`	complete	30	$(92.9, -92.8)$
`faster_nofin_full_span`	complete	60	$(112.5, -31.5)$
`faster_3fin`	crashed — insufficient ghost cells near the physical boundary	49.0	$(112.3, -23.7)$
`faster_2fin`	timed out at the 72h wall-clock limit	51.5	$(110.7, -29.4)$
`faster_midfin`	crashed — same ghost-cell error as `faster_3fin`	57.2	$(112.6, -31.8)$

Table 2: Status of each case as of the latest recorded run. Source: project case index.

Note on data provenance: the committed `faster_nofin_full_span` and `faster_3fin` case READMEs still describe earlier, incomplete attempts (an out-of-memory crash at $t = 22.9$ for the former, a still-running job at $t = 29.1$ for the latter) that predate the final results in the table above; those per-case write-ups were not updated after later, further restarts. The table above reflects the most recent recorded state.

3.2 Center-of-mass trajectories

Figure 1 overlays the center-of-mass path of every case with simulation output on one coordinate frame. The plotting convention: for each case, whichever run reached furthest is shown — a same-folder restart (resumed from checkpoint) is stitched onto its earlier data as one continuous line, while a fresh restart in a new run folder only replaces the earlier line once its own reach exceeds it. The red dashed box is the fin series' computational domain ($x \in [-30, 120]$, $y \in [-120, 30]$); the purple dotted box is a smaller 100×45 reference domain ($x \in [-20, 80]$, $y \in [-27, 18]$) shown for scale.

The two bare-cylinder baseline cases, `faster_nofin` and `faster_nofin_large`, fall almost on top of each other on a steep, nearly straight path down to $y \approx -67$ to -93 before their runs end. Every other case — `faster_nofin_full_span` and all three finned cases — peels away from that steep path much earlier and curves into a much shallower descent, ending in a band around $y \approx -24$ to -39 at $x \approx 110$–113. Within that band, `faster_3fin` (orange) sits clearly highest/flattest, with `faster_2fin`, `faster_midfin`, and `faster_nofin_full_span` bunched lower and close together. That last point is worth being careful about: because all four of those cases were stopped at close to the same x (the ghost-cell boundary issue in Section 3 above), their similar endpoints may partly just reflect being cut off at a similar point along a still-evolving curve, rather than four cases that have each reached a converged glide angle.

3.3 Effect of fins on glide angle

Where glide-angle data is available, adding fins flattens the trajectory substantially. The bare-cylinder cases settle into a glide angle (measured from horizontal) of roughly 39°–42° once their trajectory stabilizes (`faster_nofin_large`, steady by $t \approx 20$). The early, uninterrupted run of `faster_3fin` (job 14817768, reaching $t = 29.1$ before later restarts) shows the glide angle dropping steadily from 48.2° at $t = 2$–3 to just 2.5° by $t = 26$–29 — the body is moving almost horizontally by the end of that run, a qualitatively different trajectory shape from the bare cylinder, and the one case where the flattening is backed by a full time series rather than just a final position. No comparably detailed glide-angle series was committed for `faster_midfin` or `faster_2fin`; the

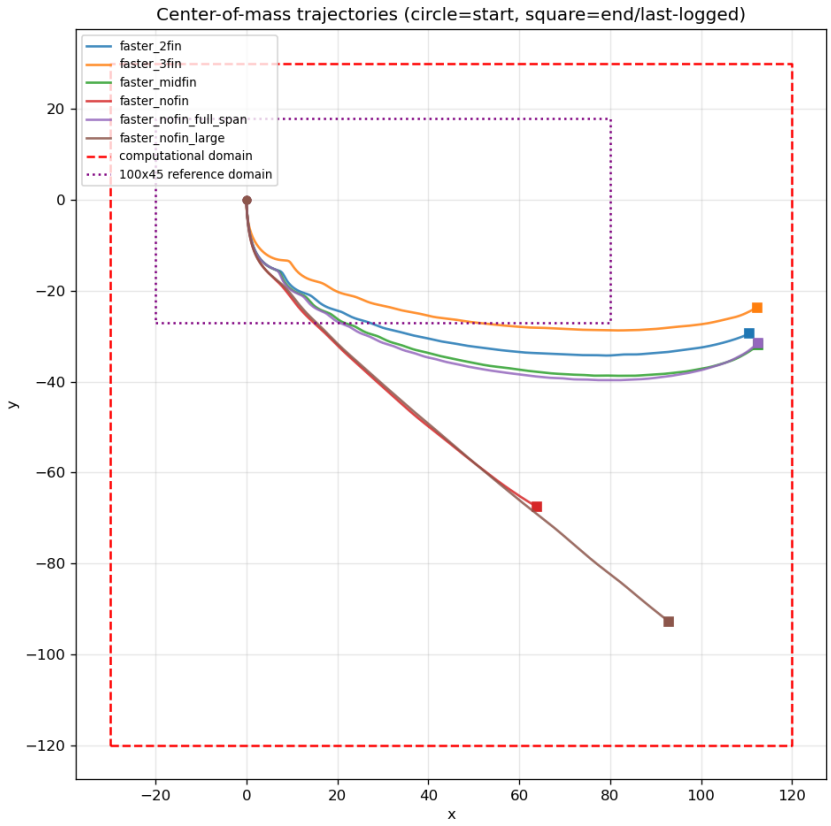


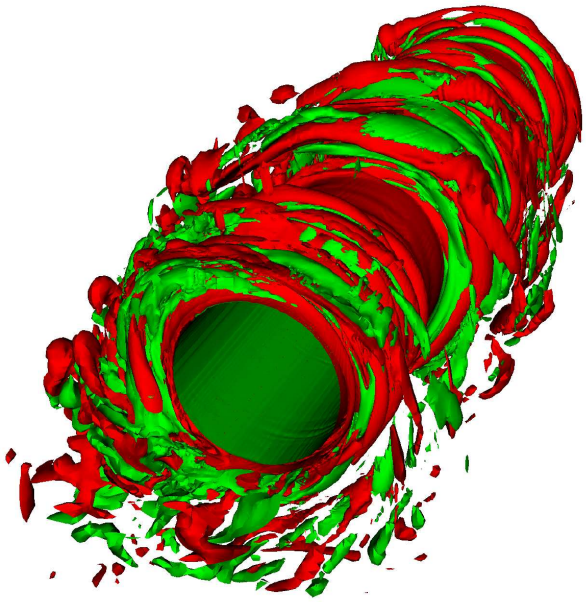
Figure 1: Center-of-mass trajectories for every case with simulation output (circle = start, square = end / last logged point). Source: project case index.

trajectory plot above (Figure 1) shows both of them also flattening relative to the bare cylinder, but — per the caveat above — not by how much more than `faster_nofin_full_span`, itself finless, does over the same stretch. So whether the flattening effect scales with fin count (1 vs. 2 vs. 3), or is mostly a 0-vs-any-fins effect, is not yet established from the current data.

3.4 Flow structure: bare cylinder vs. one fin

Figures 2 and 3 compare instantaneous vorticity ($\omega_z = \pm 1$) iso-surfaces for `faster_nofin_full_span` (no fins) and `faster_midfin` (one central fin) at $t \approx 22$, from two camera angles: looking obliquely down the body's spanwise axis (Figure 2), and looking along the streamwise direction, into the cross-flow/spanwise plane (Figure 3).

The bare cylinder sheds a comparatively regular, coiled wake that is uniform along the span – consistent with there being nothing to break the spanwise symmetry. The single fin visibly disrupts this: the vorticity surface folds and knots around the fin location, and an additional elongated, less-structured wake is thrown off along the body axis near the fin itself, where the bare cylinder has no comparable feature. This is a qualitative confirmation, at the flow-structure level, of the same effect the glide-angle numbers point to – the fin is a significant local perturbation to the wake, not a minor one.

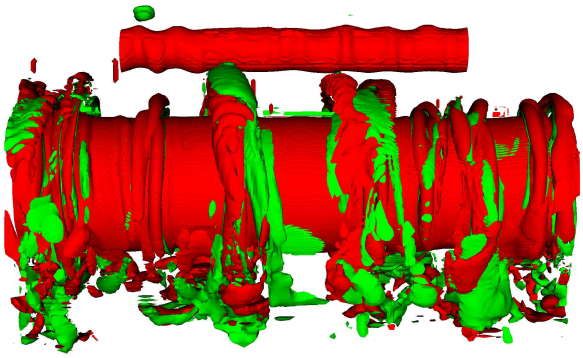


(a) `faster_nofin_full_span` (0 fins)

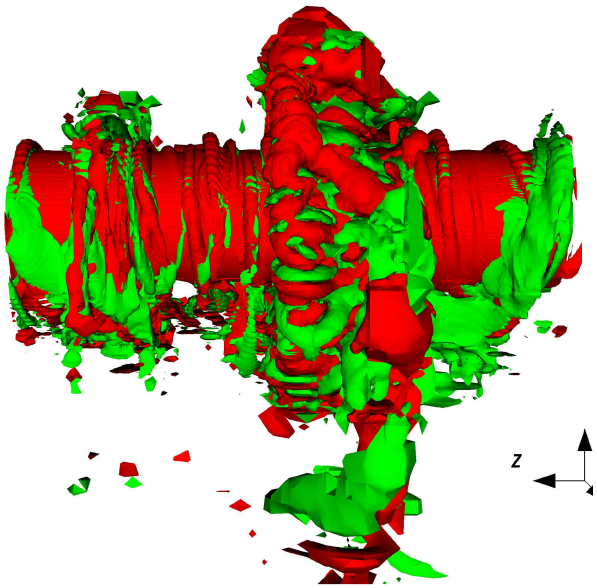


(b) `faster_midfin` (1 fin)

Figure 2: Vorticity iso-surfaces ($\omega_z = \pm 1$; red $= -1$, green $= +1$) at $t \approx 22$, viewed obliquely down the body's spanwise axis.



(a) `faster_nofin_full_span` (0 fins)



(b) `faster_midfin` (1 fin)

Figure 3: Same two cases and timestep as Figure 2, viewed along the streamwise direction (into the cross-flow / spanwise plane).

4 Open issues and next steps

- **Boundary ghost-cell crashes.** All three finned cases fail or stall near $x \approx 110$–113 from the same insufficient-ghost-cell error. A wider ghost-cell width for the IB spreading kernel, or simply stopping the run (or widening the domain) before the body reaches that region, is needed before any finned case can be compared cleanly out to $t = 30$ and beyond.
- **Incomplete glide-angle data for `faster_midfin` and `faster_2fin`.** Only `faster_3fin` has a documented glide-angle progression; repeating that analysis for 1 and 2 fins is needed to say whether the flattening effect scales with fin count or is mostly a 0-vs-any-fins effect.
- **`faster_nofin_full_span` also flattens, despite having no fins.** It ends in roughly the same y band as the finned cases (Figure 1), which could mean the fins are not the only thing driving the flattening (the longer, periodic-span body itself may matter), or could simply be an artifact of every one of these cases being cut off near the same x. Separating those two explanations needs either a full glide-angle time series for `faster_nofin_full_span` (not currently committed) or runs that get past the current boundary stopping point.
- **Stale per-case documentation.** The `faster_nofin_full_span` and `faster_3fin` READMEs describe superseded, earlier runs; they should be refreshed from the latest completed/crashed run data to avoid the kind of cross-checking this report had to do against the case index.
- **`endcaps` is not a controlled comparison.** Its differing oscillation frequency, gravity onset, and boundary condition mean any difference from `faster_nofin` cannot yet be attributed to the end caps alone.